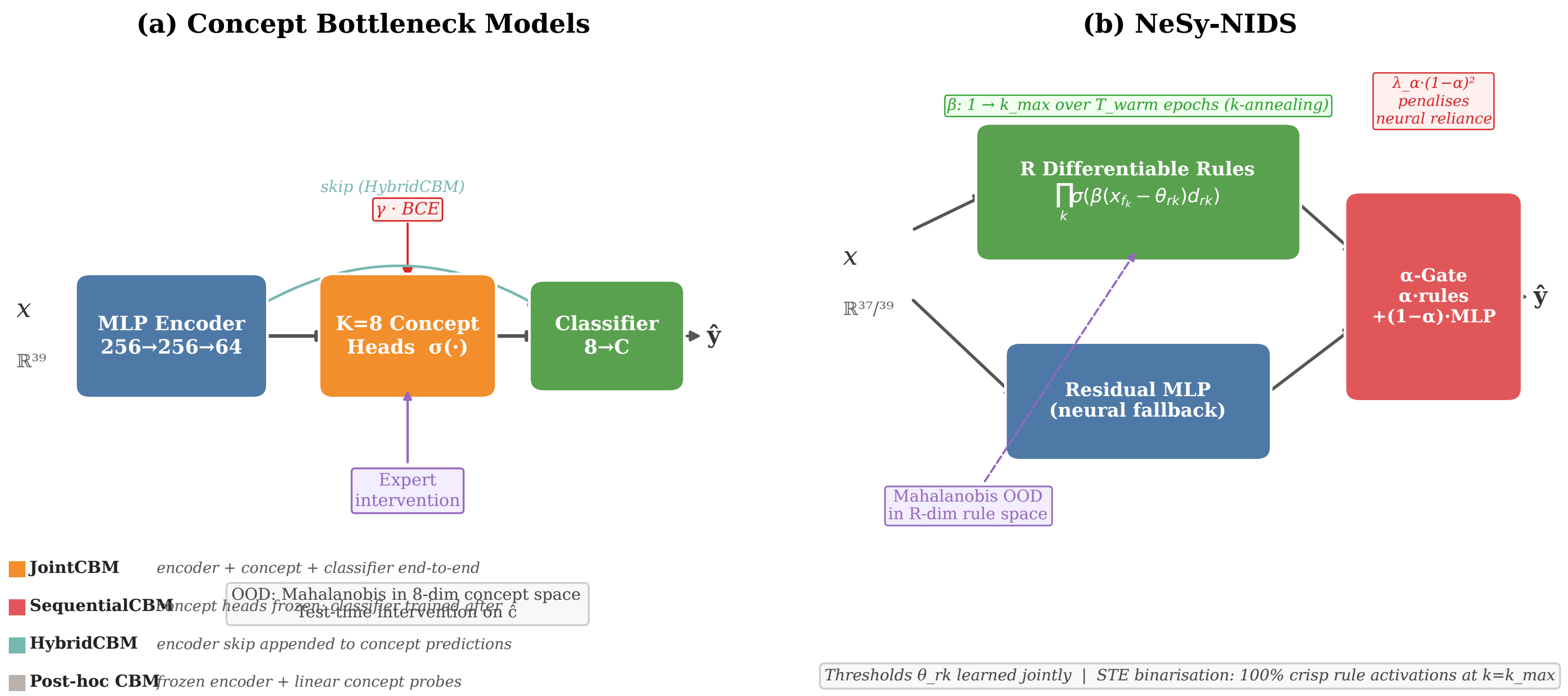

(a) Concept Bottleneck Models
x
ℝ39
MLP Encoder 256→256→64
skip (HybridCBM)
γ · BCE
K=8 Concept Heads σ(·)
Classifier 8→C
ŷ
Expert intervention
JointCBM
encoder + concept + classifier end-to-end
SequentialCBM
concept heads frozen; classifier trained after ĉ
HybridCBM
encoder skip appended to concept predictions
Post-hoc CBM
frozen encoder + linear concept probes
OOD: Mahalanobis in 8-dim concept space
Test-time intervention on ĉ
(b) NeSy-NIDS
β: 1 → k_max over T_warm epochs (k-annealing)
λ_α·(1−α)² penalises neural reliance
x
ℝ37/39
R Differentiable Rules
∏k σ(β(x_fk − θ_rk)d_rk)
Residual MLP (neural fallback)
α-Gate α·rules +(1−α)·MLP
ŷ
Mahalanobis OOD in R-dim rule space
Thresholds θ_rk learned jointly | STE binarisation: 100% crisp rule activations at k=k_max